

This exam in TMA4140 Discrete Mathematics has the following structure:



Problem 1:	Logic	10 points
Problem 2:	Sets	10 points
Problem 3:	Functions	10 points
Problem 4:	Boolean algebra	10 points
Problem 5:	Relations	10 points
Problem 6:	Induction	10 points
Problem 7:	Combinatorics	10 points
Problem 8:	Graphs and trees	10 points
Problem 9:	Number theory	10 points
Problem 10:	Finite state machines and automata	10 points
Total:		<u>100 points</u>

The answer to every problem requires a detailed argument/computation.

Problem 1 **Logic** (10 points)

- a. (3 points) Use a truth table to decide whether $[(\neg p \rightarrow q) \wedge (p \rightarrow q)] \rightarrow q$ is a tautology or a contradiction.
- b. (3 points) Write the following proposition in conjunctive normal form (CNF)

$$p \rightarrow (q \wedge \neg r).$$

- c. (4 points) Use 1st-order logic to write down the definition of surjectivity (onto) for a function $f : X \rightarrow Y$ and then form the negation of the statement so that no negation sign is to the left of a quantifier. Express the meaning of the negation in simple words.

Solution.

	p	q	$\neg p \rightarrow q$	$p \rightarrow q$	$[(\neg p \rightarrow q) \wedge (p \rightarrow q)] \rightarrow q$
a. Tautology:	0	0	0	1	1
	0	1	1	1	1
	1	0	1	0	1
	1	1	1	1	1

- b. Conjunctive normal form means that the result should be of the form $X \wedge Y$.

$$\begin{aligned} p \rightarrow (q \wedge \neg r) &\iff \neg p \vee (q \wedge \neg r) \\ &\iff (\neg p \vee q) \wedge (\neg p \vee \neg r). \end{aligned}$$

- c. Surjectivity: $\forall y \in Y \exists x \in X (f(x) = y)$,
Negation:

$$\begin{aligned} \neg(\forall y \in Y \exists x \in X (f(x) = y)) &\iff \exists y \in Y \neg(\exists x \in X (f(x) = y)) \\ &\iff \exists y \in Y \forall x \in X \neg(f(x) = y) \\ &\iff \exists y \in Y \forall x \in X (f(x) \neq y). \end{aligned}$$

Explanation in simple English: There exists an element y in the set Y such that f maps every element x in the set X to a different element than y .

Problem 2 Sets (10 points)

- a. (3 points) What is the power set of $A := \{\{a, b\}, \{c, d, e\}, \{f\}\}$?
- b. (7 points) Let A, B, C be sets. The symmetric difference is defined by $A \triangle B := (A \setminus B) \cup (B \setminus A)$. Use an element argument to show that $A \triangle B = (A \cup B) \setminus (A \cap B)$.

Solution.

- a.

$$\begin{aligned} P(A) = \{ &\emptyset, \{\{f\}\}, \{\{a, b\}\}, \{\{c, d, e\}\}, \\ &\{\{a, b\}, \{c, d, e\}\}, \{\{a, b\}, \{f\}\}, \\ &\{\{c, d, e\}, \{f\}\}, A \}. \end{aligned}$$

- b. Let $x \in (A \setminus B) \cup (B \setminus A)$. Then $x \in A \setminus B$ or $x \in B \setminus A$. If $x \in A \setminus B$, then $x \in A$ and $x \notin B$. If $x \in A$, then $x \in A \cup B$. If $x \notin B$, then $x \notin A \cap B$. Hence $x \in (A \cup B) \setminus (A \cap B)$ if $x \in A \setminus B$. If $x \in B \setminus A$, then $x \in B$ and $x \notin A$. If $x \in B$, then $x \in A \cup B$. If $x \notin A$, then $x \notin A \cap B$. Hence $x \in (A \cup B) \setminus (A \cap B)$ if $x \in B \setminus A$. Therefore $(A \setminus B) \cup (B \setminus A) \subseteq (A \cup B) \setminus (A \cap B)$.

Conversely, let $y \in (A \cup B) \setminus (A \cap B)$. Then $y \in (A \cup B)$ and $y \notin (A \cap B)$. If $y \in A \cup B$, then $y \in A$ or $y \in B$. If $y \in A$ and $y \notin A \cap B$, then $y \notin B$. Hence $y \in A \setminus B$ if $y \in A$ and $y \notin A \cap B$. If $y \in B$ and $y \notin A \cap B$, then $y \notin A$. Hence $y \in B \setminus A$ if $y \in B$ and $y \notin A \cap B$. Therefore $(A \cup B) \setminus (A \cap B) \subseteq (A \setminus B) \cup (B \setminus A)$.

We conclude that $(A \cup B) \setminus (A \cap B) = (A \setminus B) \cup (B \setminus A)$.

Problem 3 **Functions** (10 points)

- a. (3 points) Consider sets $A = \{a, b, c\}$, $B = \{1, 2, 3, 4\}$, and $C = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ and the functions

$$f_1 = \{(a, 2), (b, 3), (c, 1)\} \quad \text{and} \quad f_2 = \{(1, \alpha_2), (2, \alpha_2), (3, \alpha_4), (4, \alpha_1)\}.$$

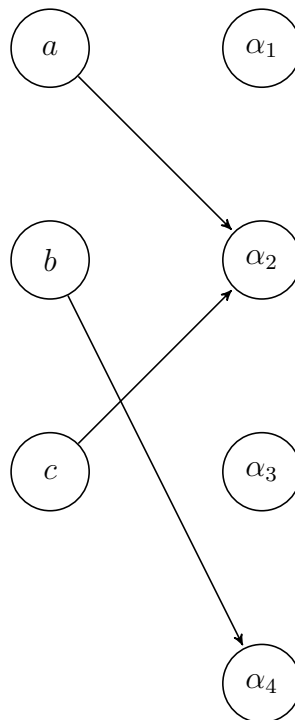
Write down the composition $f_2 \circ f_1$ as a set of ordered pairs and draw its arrow diagram.

- b. (7 points) Let m be an odd integer, that is, $m = 2l + 1$ for some integer l . Show that

$$\lfloor \frac{m^2}{4} \rfloor = \left(\frac{m-1}{2} \right) \left(\frac{m+1}{2} \right).$$

Solution.

- a. $f_2 \circ f_1 = \{(a, \alpha_2), (b, \alpha_4), (c, \alpha_2)\}.$



b.

$$\begin{aligned}
\lfloor \frac{m^2}{4} \rfloor &= \lfloor \frac{(2l+1)^2}{4} \rfloor \\
&= \lfloor \frac{4l^2 + 4l + 1}{4} \rfloor \\
&= \lfloor l^2 + l + \frac{1}{4} \rfloor \\
&= l^2 + l \\
&= l(l+1) \\
&= \frac{m-1}{2} \cdot \frac{m+1}{2}.
\end{aligned}$$

Problem 4 Boolean algebra (10 points)

Consider the notions of least common multiple (lcm) and the greatest common divisor (gcd). Let $S = \{1, 2, 3, 6\}$ and define the binary operations $x + y := \text{lcm}(x, y)$ and $x \cdot y := \text{gcd}(x, y)$ and the unary operation $\bar{x} := 6/x$. Show that $(S, +, \cdot, \bar{}, 1, 6)$ is a Boolean algebra.

Solution. Denote the Boolean algebra by B . The axioms are the following:

Commutativity:	$(\forall x, y \in B)(x + y = y + x \text{ and } x \cdot y = y \cdot x)$
Identity axioms:	$(\forall x \in B)(x + 0 = x \text{ and } x \cdot 1 = x)$
Complement axioms:	$(\forall x \in B)(x + \bar{x} = 1 \text{ and } x \cdot \bar{x} = 0)$
Distributivity of multiplication over addition:	$(\forall x, y, z \in B)(x \cdot (y + z) = x \cdot y + x \cdot z)$
Distributivity of addition over multiplication:	$(\forall x, y, z \in B)(x + y \cdot z = (x + y) \cdot (x + z))$

We check the axioms:

- Commutativity follows from $\text{lcm}(x, y) = \text{lcm}(y, x)$ and $\text{gcd}(x, y) = \text{gcd}(y, x)$.
- Additive identity: $x + 1 = \text{lcm}(x, 1) = x$. Multiplicative identity: $x \cdot 6 = \text{gcd}(x, 6) = x$.
- Additive complement: $x + \bar{x} = \text{lcm}(x, 6/x) = x \cdot \frac{6}{x} = 6$, because x and $\frac{6}{x}$ does not have any non-trivial factors in common.
- Multiplicative complement: $x \cdot \bar{x} = \text{gcd}(x, 6/x) = 1$ because x and $\frac{6}{x}$ does not have any non-trivial factors in common.

- Distributivity: Note that any element in S can be written as $x = 2^i 3^j$, with $i, j \in \{0, 1\}$. Let $x = 2^{i_x} 3^{j_x}$, $y = 2^{i_y} 3^{j_y}$, $z = 2^{i_z} 3^{j_z}$, then:

$$\begin{aligned}
 x(y + z) &= \gcd(x, \text{lcm}(y, z)) \\
 &= \gcd(x, 2^{\max(i_y, i_z)} 3^{\max(j_y, j_z)}) \\
 &= 2^{\min(i_x, \max(i_y, i_z))} 3^{\min(j_x, \max(j_y, j_z))} \\
 &= 2^{\max(\min(i_x, i_y), \min(i_x, i_z))} 3^{\max(\min(j_x, j_y), \min(j_x, j_z))} \\
 &= \text{lcm}(\gcd(x, y), \gcd(x, z)) \\
 &= xy + xz,
 \end{aligned}$$

and

$$\begin{aligned}
 x + yz &= \text{lcm}(x, \gcd(y, z)) \\
 &= 2^{\max(i_x, \min(i_y, i_z))} 3^{\max(j_x, \min(j_y, j_z))} \\
 &= 2^{\min(\max(i_x, i_y), \max(i_x, i_z))} 3^{\min(\max(j_x, j_y), \max(j_x, j_z))} \\
 &= \gcd(\text{lcm}(x, y), \text{lcm}(x, z)) \\
 &= (x + y)(x + z).
 \end{aligned}$$

Problem 5 Relation (10 points)

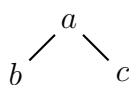
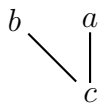
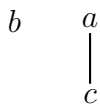
- (3 points) Draw all possible distinct Hasse diagrams for a poset with 3 elements.
- (7 points) Draw the Hasse diagram for the poset formed by the divisors of 42.

Solution.

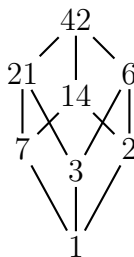
a.

$b \quad a \quad c$

$$\begin{array}{c}
 a \\
 | \\
 b \\
 | \\
 c
 \end{array}$$



b. $42 = 2 \cdot 3 \cdot 7$:



Problem 6 Induction (10 points)

For a positive integer n , define the triangular number $t_n = 1 + 2 + \cdots + n$. Use induction to show that $\sum_{i=1}^n t_i = \frac{n+2}{3}t_n$.

Hint: Recall that Carl Friedrich Gauss computed the number t_{100} as $(100 \cdot 101)/2$.

Solution. Base case: $t_1 = \frac{1+2}{3}t_1$.

Inductive step: Suppose that $\sum_{i=1}^n t_i = \frac{n+2}{3}t_n$, need to show $\sum_{i=1}^{n+1} t_i = \frac{n+3}{3}t_{n+1}$. We compute:

$$\begin{aligned}
 \sum_{i=1}^{n+1} t_i &= t_{n+1} + \sum_{i=1}^n t_i \\
 &= t_{n+1} + \frac{n+2}{3}t_n \\
 &= (n+1) + \frac{n+5}{3}t_n \\
 &= \frac{2}{3}t_n + (n+1) + \frac{n+3}{3}t_n,
 \end{aligned}$$

now we use the hint $t_n = n \cdot (n + 1)/2$ to write:

$$\begin{aligned}
 &= \frac{2}{3} \cdot \frac{n}{2} \cdot (n + 1) + (n + 1) + \frac{n + 3}{3} t_n \\
 &= \left(\frac{n}{3} + 1\right)(n + 1) + \frac{n + 3}{3} t_n \\
 &= \frac{n + 3}{3} (t_n + (n + 1)) \\
 &= \frac{n + 3}{3} t_{n+1}.
 \end{aligned}$$

Problem 7 Combinatorics (10 points)

- a. (4 points) What is the coefficient in front of the term $x^6 y^3$ in the polynomial $(3x + 4y)^9$?
- b. (6 points) We have a bin with 4 balls in different colors: red, green, brown and blue. In how many ways can we
 - i) (1 point) sample different outcomes?
 - ii) (2 points) sample different outcomes when we do not sample the red ball last?
 - iii) (3 points) sample 6 balls from the bin when we do not care about the order we sample, and we put the ball back in the bin after it is sampled?

Solution.

- a. Each term in the expansion is of the form $\binom{n}{k} \cdot 3^{n-k} \cdot 4^k \cdot x^{n-k} \cdot y^k$ where n is fixed and k goes from 0 to n . We have $n = 9$ and $k = 3$ for the term we are interested in. It follows that the coefficient is $\binom{9}{3} \cdot 3^6 \cdot 4^3 = 84 \cdot 729 \cdot 64 = 3919104$.
- b.
 - i) There are 4 options for the first ball, 3 options for the second ball, 2 option for the third ball, and 1 option for the last ball. We have $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$ ways to sample different outcomes.
 - ii) If the red ball is not the last sample, it must be the first, second or third. In each case we have $3! = 6$ ways to sample the other three balls, so we have $3 \cdot 6 = 18$ different outcomes when the red is not last.
 - iii) This is an unordered sampling with repetition with $n = 4$ balls and $k = 6$ samples. This can be visualized with 3 walls (|), for example as *red*|*green*|*brown*|*blue*, and 6 stars (*), and we get samples of the form *|*|*|*|*|* or *|*|*|*|*|* and so on. There are $\binom{n+k-1}{k}$ possible outcomes, and we get that there is $\binom{9}{6} = \frac{9!}{6! \cdot 3!} = \frac{9 \cdot 8 \cdot 7}{6} = 84$ distinct combinations.

Problem 8 **Graphs & trees** (10 points)

- a. (4 points) Let G be a graph with vertices $V = \{1, 2, 3, 4, 5\}$. Draw the graph of the following adjacency matrix, where an index (i, j) in the matrix is 0 if there is no edge from vertex i to vertex j and 1 if there is an edge. Is G directed or undirected? Is G a simple graph or a multigraph?

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- b. (6 points) Does the graph have
- i) (1 point) an Euler trail?
 - ii) (1 point) an Euler circuit?
 - iii) (2 points) a Hamiltonian path?
 - iv) (2 points) a Hamiltonian cycle?

Solution.

- a. See drawing in the figure. The graph G is directed since the matrix is not symmetric, and hence, some edges are only going from vertex i to vertex j but not from vertex j to vertex i . The graph is simple since there are no loops and no parallel edges, which we can see from the fact that all of the elements on the diagonal is 0 and that no entry is greater than 1.
- b.
- i) No, for example because vertex 1 has 4 outgoing edges but no incoming edges, so it is impossible to cross all edges in one trail.
 - ii) No, of the same reason. This can also be seen since each vertex needs equal number of in and out edges, which we do not have.
 - iii) Yes. But how do we find it? All edges connected to 1 goes out, so this must be the starting point in case a Hamiltonian path exists. All edges connected to 5 goes in, so this must be the final vertex. There is no edge from 2 to 3 and no edge from 4 to 2 or 3, and hence, the only possible Hamiltonian path is $1 \rightarrow 3 \rightarrow 2 \rightarrow 4 \rightarrow 5$.
 - iv) No. There is only one possible Hamiltonian path, and it goes from 1 to 5, but there is no edge between 5 and 1, so there can not exist any Hamiltonian cycles.

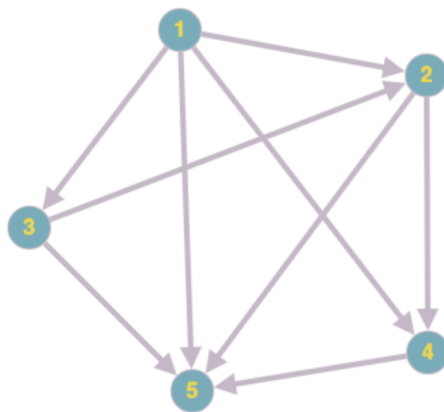


Figure 1: The graph given by the adjacency matrix in problem 8.

Problem 9 **Number theory** (10 points)

- a. (4 points) What does Euler's theorem say? Compute $134^{36362} \pmod{675}$.
- b. (6 points) Find all integer solutions x to the following system of congruences:

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 4 \pmod{7}$$

$$x \equiv 5 \pmod{8}$$

Solution.

- a. Euler's theorem says that if a and n are relatively prime positive integers then $a^{\varphi(n)} \equiv 1 \pmod{n}$, where φ is Euler's phi-function saying how many positive integers smaller or equal to n that are relatively prime to n . We observe that 675 is divisible by small primes and find the factorization $675 = 3^3 \cdot 5^2$. We also observe that 134 is not divisible by 3 or 5. Then, $\varphi(n) = (3^3 - 3^2)(5^2 - 5) = 360$ and, furthermore, $134^{360} \equiv 1 \pmod{675}$ and we have that $36362 = 101 \cdot 360 + 2$. Solution: $134^{36362} \equiv 134^2 = 17956 \equiv 406 \pmod{675}$.

- b. All moduli are relatively prime, so there exists a unique solution modulo $3 \cdot 5 \cdot 7 \cdot 8 = 840$. The solution is $x \equiv \sum_{i=1}^r a_i M_i k_i \pmod{M}$ where r is the number of congruences of the form $x \equiv a_i \pmod{m_i}$, $M = m_1 \cdot \dots \cdot m_r$ is the product of all individual moduli, $M_i = M/m_i$, and k_i is the inverse of M_i modulo m_i . We compute $k_1 \equiv 280^{-1} \equiv 1 \pmod{3}$ and $k_2 \equiv 168^{-1} \equiv 3^{-1} \equiv 2 \pmod{5}$ and $k_3 \equiv 120^{-1} \equiv 1 \pmod{7}$, and $k_4 \equiv 105^{-1} \equiv 1 \pmod{8}$. Solution: $x \equiv 2 \cdot 280 + 3 \cdot 168 \cdot 2 + 4 \cdot 120 + 5 \cdot 105 = 2573 \equiv 53 \pmod{840}$.

Problem 10 Finite state machines & automata (10 points)

- a. (4 points) Draw the transition diagram of the finite state machine F with input $I = \{a, b, c\}$, output $O = \{0, 1\}$ and states $S = \{s_0, s_1, s_2\}$ (initial state is s_0) and the following transition table. What is the output for the input string $aabbcc$?

F	η			μ		
	a	b	c	a	b	c
s_0	s_0	s_1	s_2	0	1	0
s_1	s_1	s_1	s_0	1	1	1
s_2	s_2	s_1	s_0	1	0	0

- b. (6 points) Draw the transition diagram of an automaton with five states and input $I = \{a, b\}$ accepting strings containing the sub-string: $abba$.

Solution.

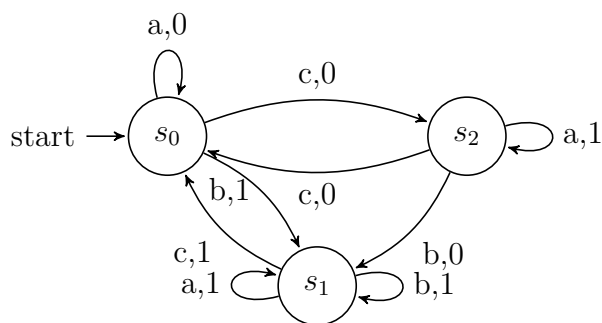


Figure 2: Solution to Problem 10.a. Output for the string $aabbcc$ is 001110.

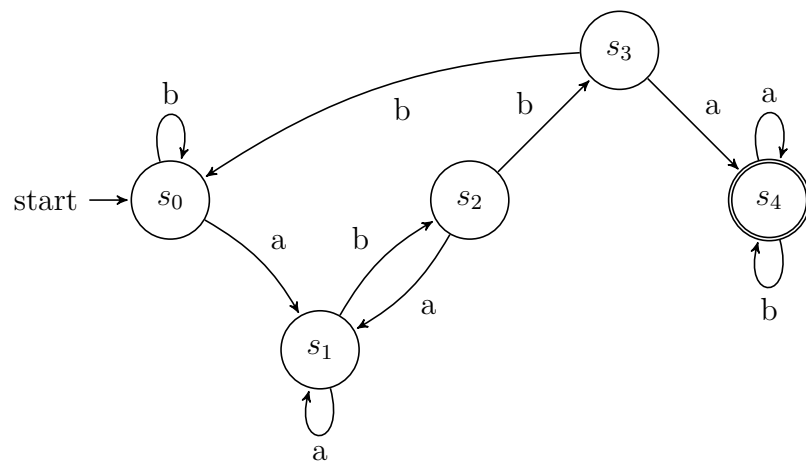


Figure 3: Solution to Problem 10.b.